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Small planar monopole ultra-wideband antenna with reduced ground plane effect

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Abstract: A printed monopole ultra-wideband antenna with reduced ground plane effect is presented. The ground plane dependence of the antenna is reduced by cutting a slot in the radiator, and then introducing a strip in the slot, to reduce the current distribution on the ground plane at a desired frequency. At this frequency, the impedance performance of the antenna is mainly affected by the monopole radiator, thus reducing the effect of the ground plane on the antenna. Measurement results confirm that this approach works well and also show good far-field properties of the antenna. The antenna is printed on a small FR4 substrate with an area of 25 mm^2; it achieves an impedance bandwidth of 3.6 to over 11 GHz, and broadband ground plane effect suppression around the design frequency of 5 GHz.

1 Introduction

The ultra-wideband (UWB) system has gained a lot of attention since the Federal Communications Commission released its frequency band in 2002 between 3.1 and 10.6 GHz. Since then, numerous types of antennas have been proposed to cover this frequency band. The planar monopole [1–6] has attracted most attention because of its low profile, small size and its ability to be integrated with other circuit components in a wireless device or radio frequency (RF) circuit. Previous works show that the effect of the ground plane on planar monopole antennas cannot be over-estimated. It has been shown that the ground plane has significant effects on antenna properties such as impedance bandwidth, radiation patterns, gain and time-domain response [7–10]. This is due to the fact that the radiator does not only radiate as desired, but the currents distributed on the ground plane makes the ground plane also introduce extra resonant modes and change the antenna's current distribution, and therefore distorts its performance. This problem brings about design complexity and difficulty in deployment when the antenna has to be integrated with other circuit components. This effect can also be observed when measuring the antenna; here, the currents on the ground plane leak into the measurement cable and consequently, the measurement cable also radiates and in turn introduces errors in the measured performance of the antenna, leading to large differences between simulation and measurements.

Solutions were proposed in [11, 12]. In [11], a rectangular notch was made vertically on the radiator and an asymmetric strip was attached to the radiator to reduce current distribution on the ground plane and concentrate the currents on the radiator. However, this approach is not a general approach and it may modify the radiation patterns because of the asymmetric nature of the antenna. Another approach was introduced in [12]. A method of cutting large L slots on the top edge of the ground plane to confine the current distribution around the L slots on the ground plane in order to reduce currents on other parts of the ground plane. This approach, however, still concentrates currents on the ground plane and not on the radiator, hence making the ground plane a major contributor to the antenna's radiation performance. Moreover, very high cross-polarisation was achieved because of the presence of currents distributed around the large slots on the ground plane. Additionally, a relatively large antenna and ground plane size were used (32×35 mm^2), hence making the antenna naturally less susceptible to ground plane effect.

In this work, a technique to suppress ground plane effect on the performance of a small printed monopole antenna is presented. The antenna has the same small size as used in [11]. The antenna is designed to reduce ground plane effect around 5 GHz, like in [12]. By inserting a strip into a slot cut on the monopole, the majority of the antenna current is distributed on the radiator at this frequency. Simulated and measurement results show that the proposed technique works well. The proposed technique provides a general solution, which is easy to implement, provides broadband ground plane reduction around 5 GHz and does not degrade impedance performance unlike [11, 12].

2 Antenna geometry and design

Fig. 1 shows the geometry of the proposed antenna along with the fabricated prototype connected to the RF cable. The antenna is etched on an FR4 substrate of thickness 1.6 mm and relative permittivity of 4.4. It consists of a rectangular radiator fed by a 50 Ω microstrip line. The physical size of the antenna is $25 \times 25 \times 1.6$ mm^3. The size of the antenna is the same as that used in [11] and a lot smaller than in [12]. The size is small to ensure that it is susceptible to ground plane effect. The bevelled and staircase notches on the ground plane and radiator, respectively, have been introduced to minimise coupling between the edges of the ground plane and radiator to improve the impedance matching ($S_{11} < -10$ dB), especially at high frequency. The notched steps have the same dimensions, and both bevelled shapes have the same dimensions as well. From simulation to measurements, the optimised dimensions of the proposed antenna are shown in Fig. 1. The antenna is printed on the *xoy*-plane.

3 Theory and parametric study

Owing to the novel nature of the technique used, parametric analysis is necessary to provide antenna designers knowledge about the various elements and how they affect the antenna. Here, the parameters L_g, L_s, s and d have been mainly experimented. It

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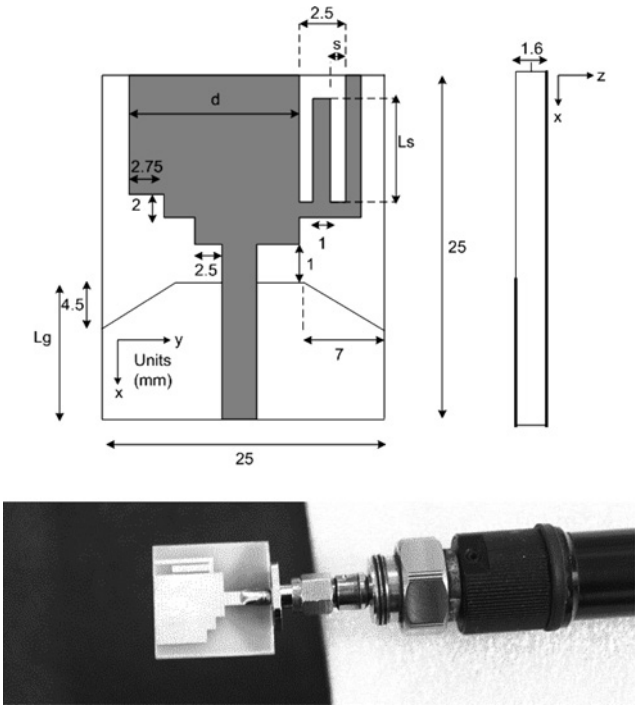


Fig. 1 *Geometry of proposed antenna and fabricated prototype attached to RF measurement cable*

should be noted that only one parameter is changed at a time while all others remain constant for each study.

3.1 Effect of antenna with/without proposed technique

Fig. 2 shows the effect of the proposed technique on the antenna. One antenna includes the slot and attached strip (proposed) shown in Fig. 1 and another without it. The conventional antenna (without slot/strip) is termed as Ant-1 and the proposed antenna is termed Ant-2 for simplicity. All other parameters are same for both antennas. It can be noted in Fig. 2 that Ant-2 shows a sharp change (resonance) in the return loss at 5 GHz whereas the Ant-1 does not. This is the effect of the technique introduced. By studying the current distribution shown in Fig. 3, it can be noted that majority of the current is distributed around the strip and edges of the slot at 5 GHz. This shows that the impedance matching at 5 GHz is more sensitive to the slot and strip dimensions than the shape and size of the ground plane. This is

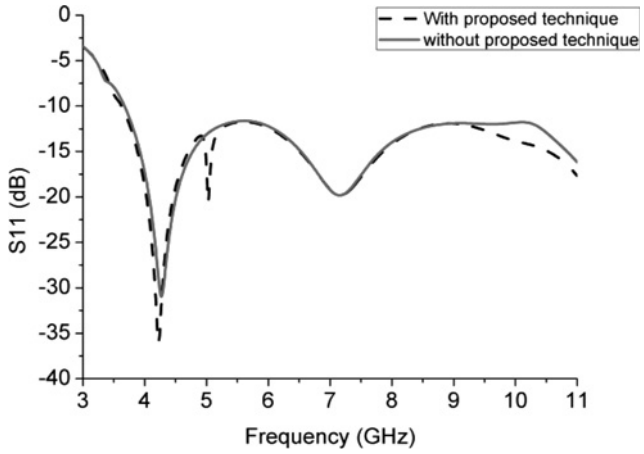


Fig. 2 *Simulated S_{11} showing the effect of antenna with/without the proposed technique*

because the current distribution on the ground plane is much weaker than those on the radiator. This means that the effect of the ground plane and the RF cable on the antenna's performance is greatly suppressed. At other frequencies, most of the currents are distributed on the feed line and edges of the ground plane, and consequently, the ground plane mostly contributes to the impedance matching at these frequencies, which is undesirable.

For the proposed technique to work, the length of the strip is about a quarter of the guided wavelength at the desired frequency. Therefore the length of the strip is calculated as

$$L_s \simeq \lambda_g/4 \tag{1}$$

$$\lambda_g = \frac{C}{f\sqrt{\varepsilon_{ff}}} \tag{2}$$

$$\varepsilon_{ff} \simeq \frac{\varepsilon_r + 1}{2} \tag{3}$$

where C is the speed of light, f is the desired frequency and ε_{ff} is the effective relative permittivity. The value of L_s in (1) can serve as a starting point to calculate the length of strip from which it can be optimised to obtain a final value. With f chosen at 5 GHz, the initial value calculated was 9 mm, and the optimised final value was 8 mm.

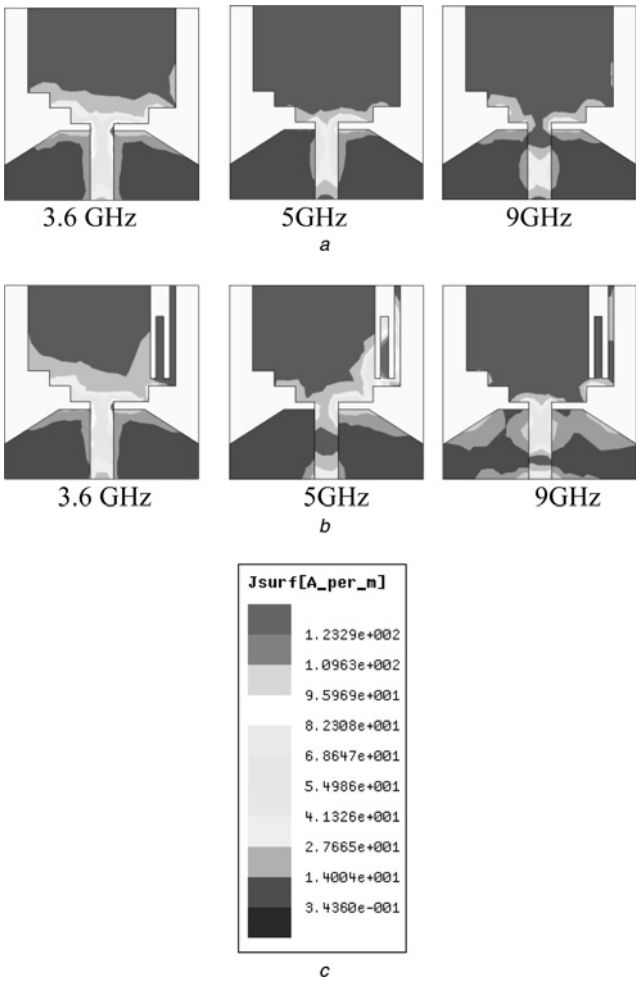


Fig. 3 *Current distribution at 3.6, 5 and 9 GHz*

a Conventional (without slot/strip) (Ant-1)
b Proposed (Ant-2)
c Colourkey

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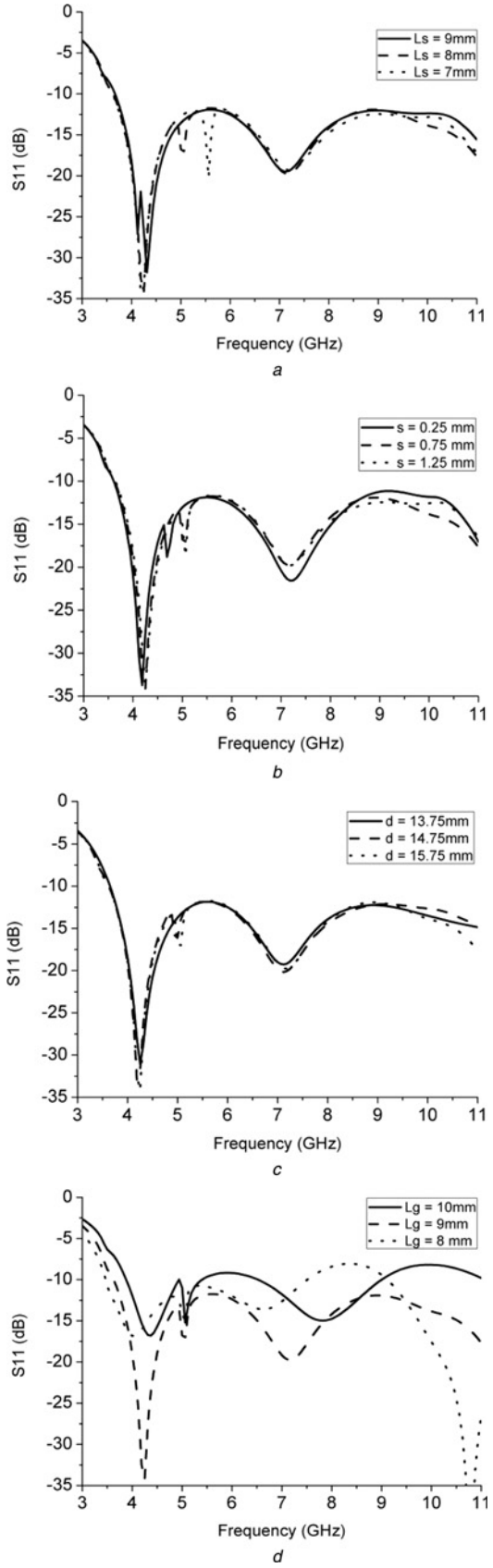


Fig. 4 *Effects of return loss on varying various parameters*

a L_s
b *s*
c *d*
d L_g

3.2 Effect of strip length, L_s

As explained in the previous section, the length of the strip is key to achieving good ground plane effect performance at a specific frequency. Fig. 4*a* shows the effect that the strip length has on the impedance matching when varied. As expected from (1), the frequency of the resonance dip shifts to a lower frequency when the length increases. It is also noted to slightly affect the intensity of the dip. Slight changes in the impedance matching are also noted at high frequency, whereas at lower frequencies the impedance matching remains almost identical.

3.3 Effect of gap, s, between strip and radiator edge

The effect of the gap, *s*, between the strip and the radiator edge is shown in Fig. 4*b*. It is noted that as the gap reduces from 0.25 to 0.75 mm, the resonance point shifts to a slightly higher frequency, where it remains unchanged when the gap further increases to 1.25 mm. Slight changes in the impedance matching are also noted at high frequency.

3.4 Effect of notch and strip position, d, on the radiator

Another important parameter to be considered is the position of the proposed technique on the radiator. Fig. 4*c* shows different positioning of the slot/strip on the radiator. The position has being varied from the 13.75 to 15.75 mm, with 15.75 and 13.75 mm being the farthest and closest from the right edge of the monopole, respectively. It is noted that as the slot and strip position moves away from the right edge of the radiator towards the middle of the radiator, the slot/strip starts to lose its effect on the antenna and the resonant point is seen to decrease. At 13.75 mm, its effect is completely lost and the Ant-1 resembles the performance in Ant-2. This is because most of the current is distributed at the edges of the radiator with little to no currents available in the centre of the radiator. Therefore by moving the slot/strip towards the middle, it has no effect on the current distribution since little or no currents are available for resonance. Therefore it is of great importance to place the slot/strip in a location around the radiator with high current density. This shows that the proposed technique is not independent of the resonant strip and slot, but involves the entire radiator.

3.5 Effect of ground plane height, L_g

Unlike the other parameters discussed, the ground plane length L_g has tremendous effect on the impedance bandwidth of the antenna at all frequencies, except at 5 GHz. This once again proves that the proposed technique is very stable and not dependent on other elements of the antenna. When the ground plane height is small, a better impedance matching is achieved at low frequency and a worse impedance matching is achieved at high frequency, and vice versa when the ground plane height is increased. This phenomenon will also be noted during measurements when the antenna is connected to the RF cable. When the antenna is attached to the RF cable during measurement, the leakage of currents from the ground plane to the RF cable will make the RF cable radiate and in effect, will make the ground plane seem 'bigger' than it actually is. Therefore the measurement results at high frequency will be similar to the parametric results in Fig. 4*d* when the ground plane length is increased. This shows that the antenna is undesirably ground plane dependent at high frequency.

For our purposes, we chose a ground plane height that gives a starting frequency of 3.6 GHz. However, Fig. 4*d* also shows that the bandwidth can be further increased to 3.1 GHz if deemed necessary. However, this is not the focus of this work.

With the knowledge of how various parameters affect the antenna, final values of L_g, L_s, *s* and *d* were chosen as 9, 8, 0.75 and 15.75 mm, respectively. The experimental results are discussed in Section 4.

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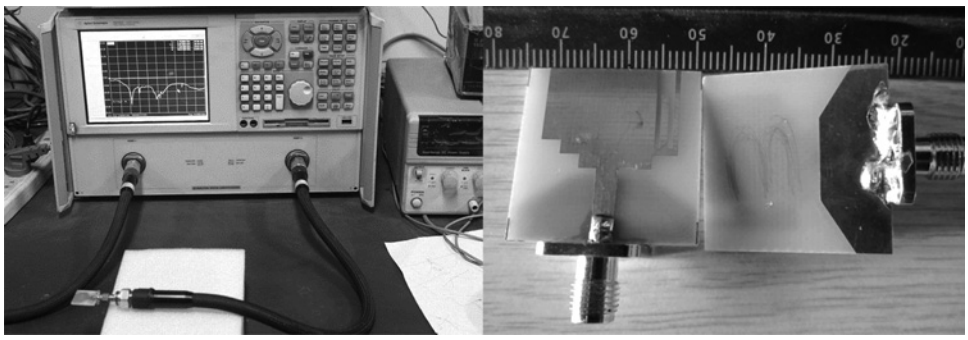


Fig. 5 *Photograph of fabricated antenna and PNA measurement setup*

4 Results and discussion

The proposed antenna is fabricated and measured with an Agilent N5230A (performance network analyser) PNA-L and a SATIMO antenna measurement system. The fabricated antenna is shown in Fig. 5. The combined simulated and measured S_{11} plot is shown in Fig. 6. The experimental results show that the antenna can cover bandwidths of 3.6–11 GHz for voltage standing wave ratio (VSWR) ≤2. Around 9 GHz, the measured S_{11} is noted to be around VSWR ≤2.5 GHz. Moreover, there is quite a discrepancy between the measured and simulated results around 9 GHz. This is because, at this frequency, the antenna is ground plane dependent since most of the currents are distributed on the ground plane with minimal currents on the radiator, as shown in Fig. 3*b*. However, at the frequency of interest, 5 GHz, the measured S_{11} is noted to match perfectly with the simulated one (as shown in the dotted ellipse). This result is in accordance with our theory and confirms that the ground plane effect is suppressed at 5 GHz as calculated. Even better, the impedance matching is also noted to be in perfect agreement around 4–6 GHz showing broad band ground plane effect suppression.

To further prove the feasibility of the proposed technique, the conventional antenna, that is, the same antenna without the proposed technique is also measured and compared with the proposed technique in terms of impedance matching, far field properties and time-domain characteristics. Fig. 7 shows the S_{11} plot of the conventional antenna. It can be seen that there is quite a discrepancy between the simulated and measured results around 5 GHz. Compared with Fig. 6, this discrepancy is removed because of the proposed technique and therefore the ground plane effect is suppressed around this frequency.

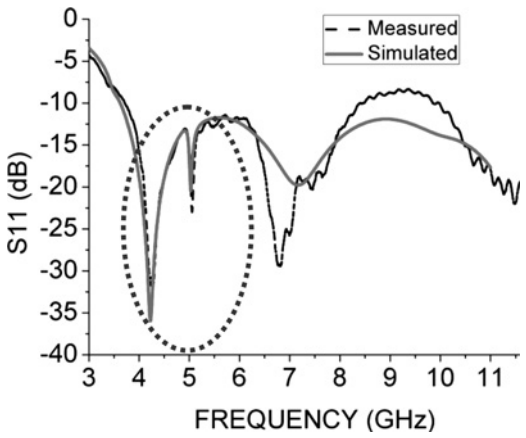


Fig. 6 *Measured and simulated return loss of the proposed antenna*

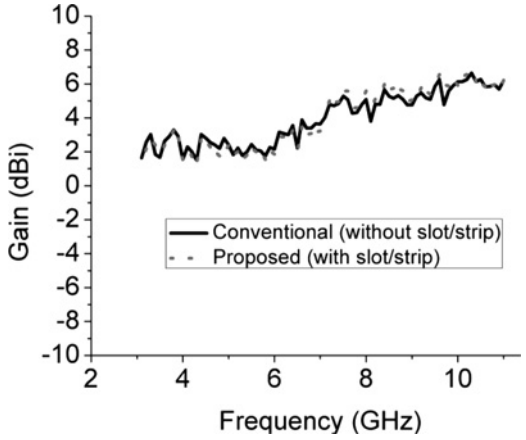


Fig. 8 *Measured peak gains of the proposed and conventional antennas*

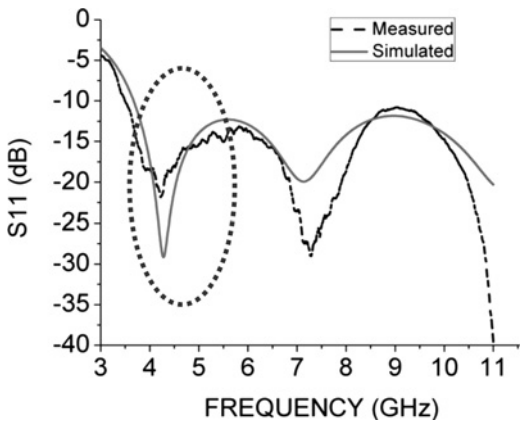


Fig. 7 *Measured and simulated return loss of the conventional antenna*

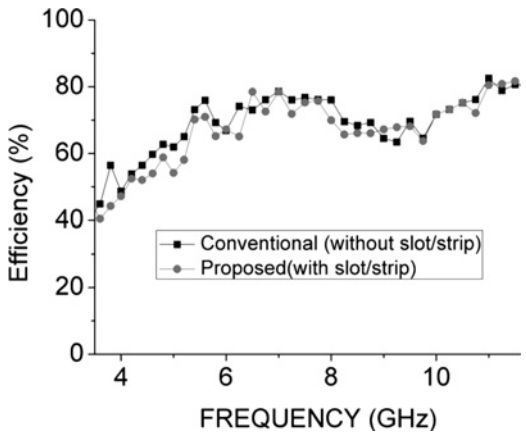
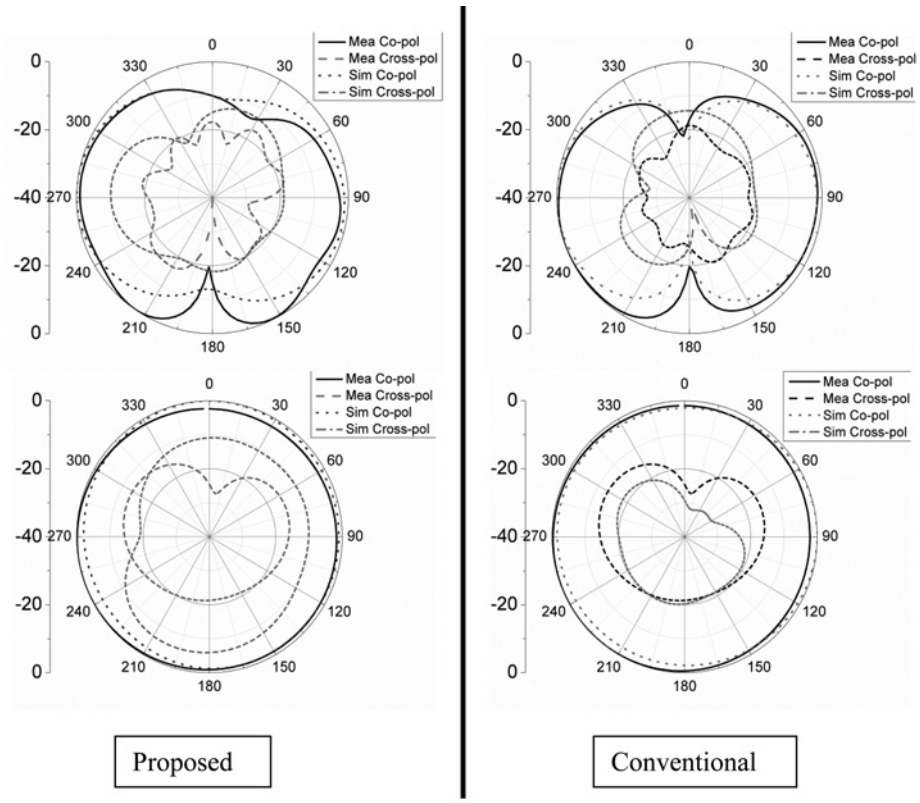
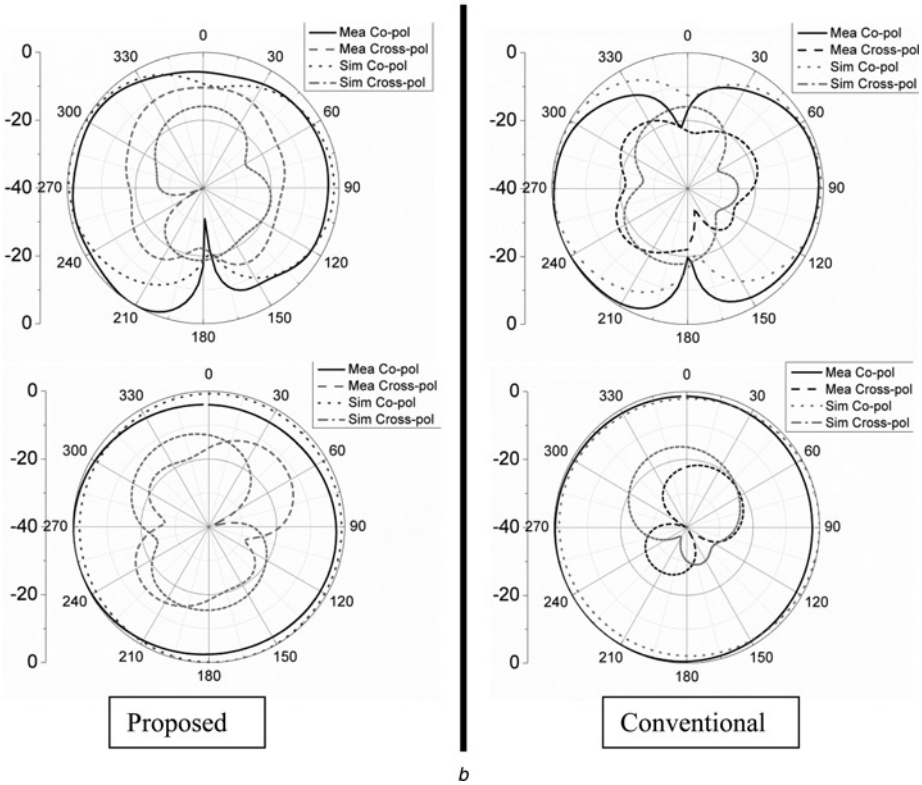


Fig. 9 *Measured efficiency of the proposed and conventional antennas*

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a



b

Fig. 10 *Measured E-planes (top) and H-planes (bottom) radiation patterns at*

a 3.6 GHz
b 5 GHz
c 6 GHz

For the far field results, the peak gains of the proposed and conventional antennas have been plotted together as shown in Fig. 8. It shows that the peak gains of the both antennas are almost identical showing that the proposed technique does not affect the performance of the antenna. An average peak gain of 4 dBi is achieved within the operating band. The measured

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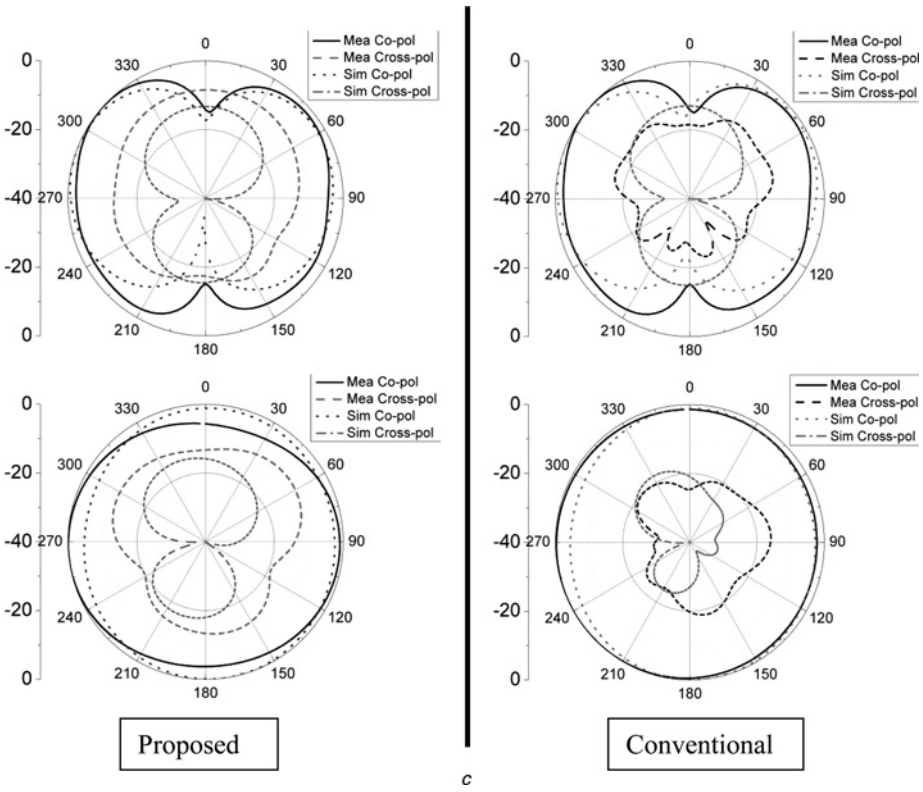


Fig. 10 *Continued*

efficiencies are also shown in Fig. 9. An average efficiency of 65% is recorded across the operating bandwidths. Fig. 10 shows the compared radiation patterns of the proposed antenna (left) and the conventional antenna (right). In both cases, the antenna is printed on the *xoy*-plane; hence the *E*-plane (*xoy*, $\theta = -90°$) and *H*-plane (*yoz*, $\phi = 90°$) radiation patterns are plotted at 3.6, 5 and 6 GHz. It can be noted that the *E*-plane radiation pattern depicts a monopole-like radiation pattern as expected and the *H*-plane depicts an omnidirectional radiation pattern as expected, both with minimal cross-polarisation. Unlike [11, 12], the proposed technique does not significantly affect the radiation patterns and good agreements between simulations and measurements are obtained. The time-domain properties of both antennas have been measured as well. The group delays are shown in Fig. 10. It shows an almost identical time-domain performance. This means the proposed technique does not distort the performance of the conventional antenna (Fig. 11). The plot shows a constant group delay curve of the antenna.

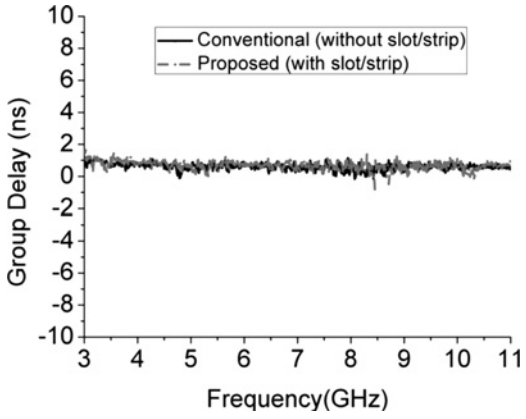


Fig. 11 *Measured group delay of the proposed and conventional antennas*

This means that the antenna can radiate and receive short pulse signals without distortion.

5 Conclusion

In this paper, a small printed antenna has been proposed for UWB applications. The antenna has been designed to reduce ground plane effects around 5 GHz. By cutting a slot on the right edge of radiator and attaching a horizontal strip to it, majority of the currents at this frequency can be concentrated on the radiator and less on the ground plane. The measured results prove that this technique makes the antenna less sensitive to the ground plane in terms of impedance matching. The simulated and measured results were identical at the design frequency thereby proving that the attached RF measurement cable had no significant effect on the antenna. The radiation pattern and time-domain properties of the proposed antenna when compared with the conventional antenna remain the same showing that the proposed technique does not distort the antenna performance. Additionally, broadband ground plane effect suppression is achieved within 4–6 GHz. This property makes the antenna attractive to practical applications in mobile devices.

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